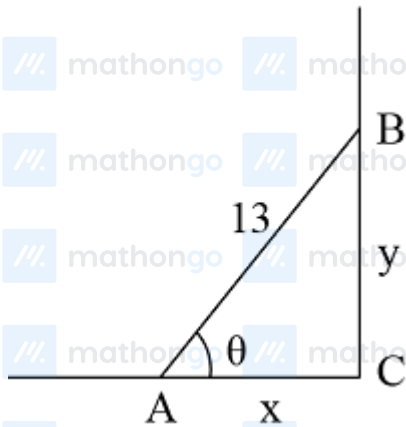


Hints & Solutions

#MathBoleTohMathonGo

Q1 (2) - Derivative as Rate Measure

Let $AB = 13\text{ft}$ is ladder and $AC = x, BC = y$.



In right angled $\triangle ACB$, $x^2 + y^2 = (13)^2 \dots(i)$

$$\Rightarrow \tan \theta = \frac{y}{x} \dots(ii)$$

It is given that, $\frac{dx}{dt} = 5\text{ft/s} \dots(iii)$

On differentiating Eq. (i) w.r.t. t , we get

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = \frac{-x}{y} \frac{dx}{dt} \dots(iv)$$

On differentiating Eq. (ii) w.r.t. t , we get

$$\begin{aligned} \sec^2 \theta \frac{d\theta}{dt} &= \frac{x \frac{dy}{dt} - y \frac{dx}{dt}}{x^2} \\ &= \frac{-x \times \frac{x}{y} \frac{dx}{dt} - y \frac{dx}{dt}}{x^2} \quad [\text{using Eq. (iv)}] \end{aligned}$$

$$\begin{aligned} \Rightarrow \sec^2 \theta \frac{d\theta}{dt} &= \frac{\left(-\frac{x^2}{y} - y\right) \frac{dx}{dt}}{x^2} \\ \Rightarrow \frac{d\theta}{dt} &= \frac{\left(-\frac{x^2}{y} - y\right) \frac{dx}{dt}}{x^2 \sec^2 \theta} \dots(v) \end{aligned}$$

When $x = 12$, then $y = \sqrt{(13)^2 - (12)^2} = 5$

$$\text{Now, } \left(\frac{d\theta}{dt}\right)_{x=12, y=5} = \frac{\left(\frac{-144}{5} - 5\right) 5}{(12)^2 \left(\frac{13}{12}\right)^2} \left[\because \sec \theta = \frac{13}{12}\right]$$

$$= \frac{\left(\frac{-144-25}{5}\right) \times 5}{(13)^2} = \frac{-169}{(13)^2} = -1\text{rad/s}$$

Q2 (2) - Derivative as Rate Measure

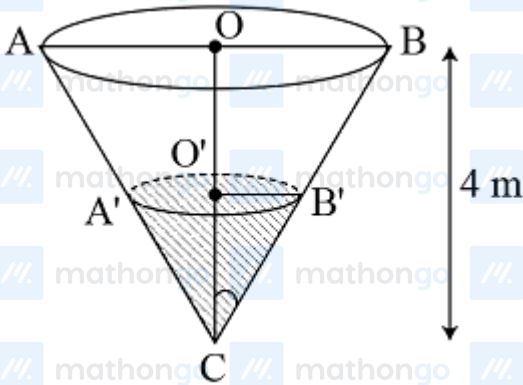
#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

Let α be the semi-vertical angle of the cone $C'AB$ whose height CO is 4 m and radius $OB = 2$ m, then

$$\tan \alpha = \frac{1}{2}$$



Let V be the volume of water in the cone i.e. the volume of the cone $CA'B'$

After time t min, h be the height of the water, then

$$V = \frac{1}{3}\pi(O'B')^2(CO')$$

$$\Rightarrow V = \frac{1}{3}\pi h^3 \tan^2 \alpha \quad \left[\because \tan \alpha = \frac{O'B'}{CO'} = \frac{O'B'}{h} \right]$$

$$\Rightarrow V = \frac{\pi}{12}h^3$$

$$\text{Now, } \frac{dV}{dt} = \frac{\pi}{12}3h^2 \frac{dh}{dt} = \frac{\pi}{4}h^2 \frac{dh}{dt}$$

$$\Rightarrow 77 \times 10^3 = \frac{\pi h^2}{4} \cdot \frac{dh}{dt} \quad \left[\begin{array}{l} \because 1 \text{ L} = 10^3 \text{ cm}^3 \\ \therefore 77 \text{ L} = 77 \times 10^3 \text{ cm}^3 \\ \text{i.e. } \frac{dV}{dt} = 77 \text{ L} = 77 \times 10^3 \text{ cm}^3 \end{array} \right]$$

$$\Rightarrow \frac{dh}{dt} = \frac{77 \times 4 \times 10^3}{\pi h^2}$$

$$\therefore \left(\frac{dh}{dt} \right)_h = \frac{77 \times 4 \times 10^3}{\pi \times (70)^2} = \frac{77 \times 4 \times 7 \times 10^3}{22 \times 70 \times 70} = 20 \text{ cm/min}$$

Q3 (3) - Derivative as Rate Measure

Given equation of the curve is

$$y^3 = 27x \dots(i)$$

On differentiating Eq. (i) w.r.t. y , we get

$$3y^2 = 27 \frac{dx}{dy} \Rightarrow \frac{dx}{dy} = \frac{y^2}{9}$$

If abscissa changes at a slower rate than the ordinate, then

Hints & Solutions

#MathBoleTohMathonGo

$$\left| \frac{dx}{dy} \right| < 1 \Rightarrow \left| \frac{y^2}{9} \right| < 1$$

$$\Rightarrow |y^2| < 9$$

$$\Rightarrow -3 < y < 3$$

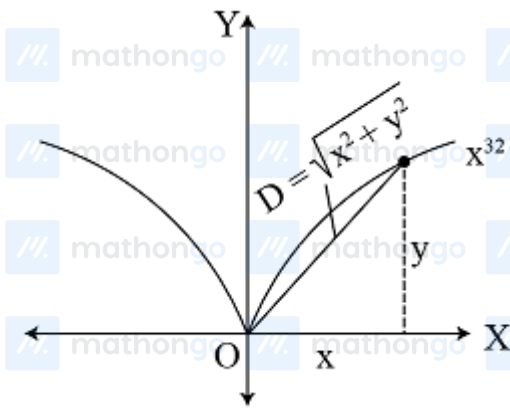
$$\therefore y \in (-3, 3) \text{ and } x \in (-1, 1)$$

Q4 (1) - Derivative as Rate Measure

Given equation of the curve is

$$y = x^{3/2}$$

$$\Rightarrow \frac{dy}{dt} = \frac{3}{2} x^{1/2} \frac{dx}{dt} \dots (i)$$



Let D be the distance of particle from origin i.e. $D = \sqrt{x^2 + y^2}$,

$$\therefore \frac{dD}{dt} = \frac{1}{2\sqrt{x^2 + y^2}} \left(2x \frac{dx}{dt} + 2y \frac{dy}{dt} \right)$$

$$\Rightarrow \frac{dD}{dt} = \frac{1}{2\sqrt{x^2 + y^2}} \left(2x \frac{dx}{dt} + 2y \frac{3}{2} x^{1/2} \frac{dx}{dt} \right)$$

$$\Rightarrow 11 = \frac{1}{2\sqrt{x^2 + y^2}} (2x + 3y^{1/2}) \frac{dx}{dt} \left[\because \frac{dD}{dt} = 11 \text{ units/s} \right]$$

$$\text{Now, } \frac{dx}{dt} = \frac{22\sqrt{x^2 + y^2}}{(2x + 3y^{1/2})}$$

$$\therefore \left(\frac{dx}{dt} \right)_{x=3} = \frac{22\sqrt{9+27}}{[2 \times 3 + 3 \times (3)^{1/2}]} \left[\begin{array}{l} \text{when } x = 3, \\ \text{then } y = (3)^{3/2} \end{array} \right]$$

$$= \frac{22 \times 6}{33} = \frac{132}{33} = 4$$

Q5 (3) - Derivative as Rate Measure

Given, x and y are the sides of squares.

Then, the area of the squares are x^2 and y^2

$$\text{We have to obtain, } \frac{d(y^2)}{d(x^2)} = \frac{2y \frac{dy}{dx}}{2x} = \frac{y}{x} \times \frac{dy}{dx} \dots (i)$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

Given curve is $y = x - x^2$

$$\Rightarrow \frac{dy}{dx} = 1 - 2x \dots (ii)$$

Thus, $\frac{d(y^2)}{d(x^2)} = \frac{y}{x} (1 - 2x)$ [from Eqs. (i) and (ii)]

$$\Rightarrow \frac{d(y^2)}{d(x^2)} = \frac{(x-x^2)(1-2x)}{x}$$

$$\Rightarrow \frac{d(y^2)}{d(x^2)} = 2x^2 - 3x + 1$$

$$\therefore \left[\frac{d(y^2)}{d(x^2)} \right]_{x=2} = 2 \times 2^2 - 3 \times 2 + 1 = 3$$

Q6 (4) - Tangents and Normals

Given, $y = x + \sin x \cos x$

Now, $\frac{dy}{dx} = 1 + \cos^2 x - \sin^2 x$

$$\therefore \left(\frac{dy}{dx} \right)_{x=\frac{\pi}{2}} = 0$$

When $x = \frac{\pi}{2}$, then $y = \frac{\pi}{2}$

$\therefore$ Equation of the normal at $\left(\frac{\pi}{2}, \frac{\pi}{2} \right)$ is

$$y - \frac{\pi}{2} = \frac{-1}{0} \left(x - \frac{\pi}{2} \right)$$

$$\Rightarrow x = \frac{\pi}{2} \Rightarrow 2x = \pi$$

Q7 (3) - Tangents and Normals

Given, $x + y = x^y \dots (i)$

Since, the curve cuts X -axis, where $y = 0$.

$$\therefore x = x^0 \Rightarrow x = 1$$

So the point of contact is $(1, 0)$

On differentiating Eq. (i) w.r.t. x , we get

$$1 + \frac{dy}{dx} = x^y \left(\frac{y}{x} + \log x \cdot \frac{dy}{dx} \right)$$

$$\Rightarrow 1 + \frac{dy}{dx} - x^y \frac{dy}{dx} \cdot \log x = x^y \cdot \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} (1 - x^y \log x) = x^y \frac{y}{x} = -1$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^y \left(\frac{y}{x} \right) - 1}{(1 - x^y \log x)}$$

$$\therefore \left(\frac{dy}{dx} \right)_{(1,0)} = \frac{-1}{1} = -1$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

So, the equation of normal at $(1, 0)$ is

$$y - 0 = 1(x - 1) \Rightarrow x - y - 1 = 0$$

Q8 (2) - Tangents and Normals

Given, $\frac{x^4}{2} = x + y \dots(i)$

Let (x_1, y_1) be the point on the curve (i), then

$$\frac{x_1^4}{2} = x_1 + y_1 \dots(ii)$$

Also, point (x_1, y_1) lies on the line $3x + 4y = c$

$$\therefore 3x_1 + 4y_1 = c \dots(iii)$$

From Eq. (i), we have

$$\begin{aligned} \frac{dy}{dx} &= 2x^3 - 1 \\ \Rightarrow \left(\frac{dy}{dx}\right)_{(x_1, y_1)} &= 2x_1^3 - 1 \end{aligned}$$

Now, slope of tangent at $(x_1, y_1) = \text{Slope of the line}$

$$\Rightarrow 2x_1^3 - 1 = \frac{-3}{4}$$

$$\Rightarrow 2x_1^3 = \frac{1}{4}$$

$$\Rightarrow x_1^3 = \frac{1}{8} \Rightarrow x_1 = \frac{1}{2}$$

$$\therefore y_1 = \frac{1}{4} \left(c - \frac{3}{2} \right)$$

From Eq. (ii), we have

$$\left(\frac{1}{2}\right)^4 = \frac{1}{2} + \frac{1}{4} \left(-\frac{3}{2} + c \right)$$

$$\frac{1}{32} = \frac{1}{2} - \frac{3}{8} + \frac{c}{4} \Rightarrow \frac{1}{32} = \frac{1}{8} + \frac{c}{4}$$

$$\Rightarrow \frac{c}{4} = \frac{1}{32} - \frac{1}{8} \Rightarrow \frac{c}{4} = \frac{-3}{32}$$

$$\therefore c = \frac{-3}{8}$$

Hence, only one value of c is possible.

Q9 (2) - Tangents and Normals

Given, $f(x) = x^3 + ax + b, a \neq b \dots(i)$

From Eq. (i), we have

$$f'(x) = 3x^2 + a$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$f'(a) = 3a^2 + a$$

$$f'(b) = 3b^2 + a$$

According to the question,

$$f'(a) = f'(b)$$

$$3a^2 + a = 3b^2 + a$$

$$\Rightarrow 3a^2 = 3b^2 \Rightarrow a = \pm b$$

$$\Rightarrow a = -b$$

$$[\because a \neq b]$$

$$\text{Now, } f(1) = 1 + a + b = 1 - b + b \quad [\because a = -b]$$

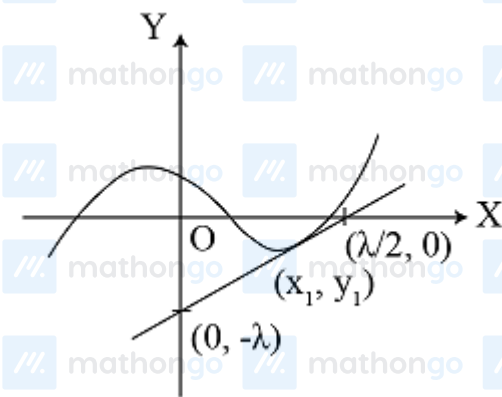
$$\therefore f(1) = 1$$

Q10 (2) - Tangents and Normals

Here, slope of the tangent is

$$\frac{0+\lambda}{\lambda/2-0} = 2 \dots (i)$$

$$\text{and } y = x^3 - 3x^2 - 7x + 6$$



$$\left(\frac{dy}{dx} \right)_{(x_1, y_1)} = 3x_1^2 - 6x_1 - 7 \dots (ii)$$

From Eqs. (i) and (ii), we get

$$3x_1^2 - 6x_1 - 7 = 2$$

$$\Rightarrow 3x_1^2 - 6x_1 - 9 = 0$$

$$\Rightarrow x_1^2 - 2x_1 - 3 = 0$$

$$\Rightarrow (x_1 - 3)(x_1 + 1) = 0$$

$$\Rightarrow x_1 = 3 \quad [\text{neglecting } x_1 = -1]$$

Hints & Solutions

#MathBoleTohMathonGo

Also, $y_1 = x_1^3 - 3x_1^2 - 7x_1 + 6 = -15$

$\therefore$ So, the required point is $(3, -15)$.

Q11 (2) - Tangents and Normals

Given, $y = x^n (n \in \mathbb{N})$

On differentiating. W.r.t. x , we get

$$\frac{dy}{dx} = nx^{n-1}$$

$$\Rightarrow \left(\frac{dy}{dx} \right)_{(a, a^n)} = n \cdot a^{n-1}$$

Equation of the normal at (a, a^n) is

$$y - a^n = \frac{-1}{na^{n-1}}(n - a) \dots(i)$$

Since, Eq. (i) intersect Y -axis at $(0, b)$.

$$\therefore b - a^n = \frac{a}{na^{n-1}} = \frac{1}{n} a^{2-n}$$

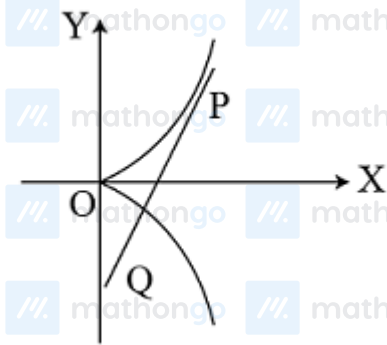
On taking limit $a \rightarrow 0$ both sides, we get

$$\lim_{a \rightarrow 0} b - \lim_{a \rightarrow 0} a^n = \frac{1}{n} \lim_{a \rightarrow 0} a^{2-n}$$

$$\Rightarrow \frac{1}{2} = \frac{1}{n} \Rightarrow n = 2 \quad \left[\because \lim_{a \rightarrow 0} b = \frac{1}{2} \right]$$

Q12 (2) - Tangents and Normals

Let $P = (x_1, x_1^{3/2})$ and $Q = (x_2, x_2^{3/2})$



Here, $y^2 = x^3$

$$\Rightarrow 2y \frac{dy}{dx} = 3x^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{3x_1^2}{2y_1} = \frac{3}{2} \frac{x_1^2}{x_1^{3/2}} = \frac{3}{2} x_1^{1/2} \dots(i)$$

$$\text{Slope of } PQ = \frac{x_2^{3/2} - x_1^{3/2}}{x_2 - x_1} \dots(ii)$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

From Eqs. (i) and (ii), $\frac{x_2^{3/2} - x_1^{3/2}}{x_2 - x_1} = \frac{3}{2} x_1^{1/2}$

$$\Rightarrow 2\left(\sqrt{\frac{x_2}{x_1}}\right)^3 - 3\left(\sqrt{\frac{x_2}{x_1}}\right)^2 + 1 = 0$$

On solving $\sqrt{\frac{x_2}{x_1}} = 1, -\frac{1}{2}$

Also, $\frac{\tan \alpha}{\tan \beta} = \sqrt{\frac{x_1}{x_2}} = -2$ or 1

Q13 (4) - Tangents and Normals

We have, $y = x + \frac{4}{x^2}$

On differentiating w.r.t. x , we get

$$\frac{dy}{dx} = 1 - \frac{8}{x^3}$$

Since, the tangent is parallel to X -axis, therefore

$$\frac{dy}{dx} = 0$$

$$\Rightarrow x^3 = 8$$

$$\therefore x = 2 \text{ and } y = 3$$

Q14 (1) - Tangents and Normals

We have,

$$y = \int_0^x |t| dt \text{ at, } x \in R \quad \dots (i)$$

$$\frac{dy}{dx} = |t|, x \in R$$

Let (x_1, y_1) be a point on the curve $y = \int_0^x |t| dt$, where tangent is parallel to the line $y = 2x$, then

$$\left(\frac{dy}{dx}\right)_{(x_1, y_1)} = 2$$

$$\Rightarrow |x_1| = 2 \Rightarrow x_1 = \pm 2$$

Since, (x_1, y_1) lies on Eq. (i).

$$\therefore y_1 = \int_0^{x_1} |t| dt$$

When $x_1 = 2$, then

$$y_1 = \int_0^2 |t| dt = \int_0^2 t dt = \left[\frac{t^2}{2}\right]_0^2 = 2$$

When $x_1 = -2$, then

$$y_1 = \int_0^{-2} -t dt = \left[\frac{-t^2}{2}\right]_0^{-2} = -2$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

So, the points on the curve are $(2, 2)$ and $(-2, -2)$. Equations of the tangent at these points are

$$\begin{aligned} y - 2 &= 2(x - 2) \text{ and } y + 2 = 2(x + 2) \\ \Rightarrow y &= 2x - 2 \text{ and } y = 2x + 2 \end{aligned}$$

Hence, the x -intercept of these tangents are ± 1 .

Q15 (1) - Tangents and Normals

Equation of tangent to the curve $y = e^x$ at (c, e^c) is

$$y - e^c = e^c(x - c) \dots(i)$$

Equation of the line joining $(c - 1, e^{c-1})$ and $(c + 1, e^{c+1})$ is

$$y - e^{c-1} = \frac{e^c(e - e^{-1})}{2}(x - c + 1) \dots(ii)$$

On subtracting Eq. (i) from Eq. (ii), we get

$$\begin{aligned} e^c - e^{c-1} &= \frac{e^c(e - e^{-1})}{2}(x - c + 1) - e^c(x - c) \\ \Rightarrow e^c - e^{c-1} &= (x - c)e^c \left(\frac{e - e^{-1}}{2} - 1 \right) + e^c \left(\frac{e - e^{-1}}{2} \right) \\ \Rightarrow x - c &= \frac{(e - 1)^2}{2 - (e - 1)^2} < 0 \end{aligned}$$

$$\Rightarrow x < c$$

Hence, Eqs. (i) and (ii) intersect at a point on the left of c .

Q16 (4) - Mean Value Theorem

$$f(x) = \begin{cases} \frac{x^3 - 2x^2 - 5x + 6}{x - 1}, & x \neq 1 \\ -6, & x = 1 \end{cases}$$

To check continuity of $f(x)$ at $x = 1$,

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} f(1 - h) \\ &= \lim_{h \rightarrow 0} \frac{(1 - h)^3 - 2(1 - h)^2 - 5(1 - h) + 6}{1 - h - 1} \\ &= \lim_{h \rightarrow 0} \frac{(1 - h)[1 + h^2 - 2h - 2 + 2h - 5] + 6}{-h} \\ &= \lim_{h \rightarrow 0} \frac{(1 - h)[h^2 - 6] + 6}{-h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 - 6 - h^3 + 6h + 6}{-h} = \lim_{h \rightarrow 0} \frac{2h - 3h^2 + 6}{-1} \\ &= -6 \end{aligned}$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

[by L' Hospital's]

$$\begin{aligned}
 \text{RHL} &= \lim_{x \rightarrow 1^+} f(x) = \lim_{h \rightarrow 0} f(1+h) \\
 &= \lim_{h \rightarrow 0} \frac{(1+h)^3 - 2(1+h)^2 - 5(1+h) + 6}{1+h-1} \\
 &= \lim_{h \rightarrow 0} \frac{3(1+h)^2 - 4(1+h) - 5}{1} \quad [\text{by L' Hospital's rule}] \\
 &= -6
 \end{aligned}$$

$$\text{Also, } f(1) = -6$$

$$\therefore \text{LHL} = \text{RHL} = f(1) = -6$$

$\therefore f(x)$ is continuous at $x = 1$ To check differentiability of $f(x)$ at $x = 1$,

$$\begin{aligned}
 Lf'(1) &= \lim_{h \rightarrow 0} \frac{(1-h)^3 - 2(1-h)^2 - 5(1-h) + 6}{-1} \\
 &[\text{by L' Hospital's rule}] \\
 &= \lim_{h \rightarrow 0} \frac{3(1-h)^2(-1) - 2 \times 2(1-h)(-1) - 5(-1)}{-1} \\
 &= \lim_{h \rightarrow 0} \frac{3(1-h)^2 - 4(1-h) - 5}{1} \\
 &= -6
 \end{aligned}$$

$$\begin{aligned}
 Rf'(1) &= \lim_{h \rightarrow 0} \frac{(1+h)^3 - 2(1+h)^2 - 5(1+h) + 6}{h} \\
 &= \lim_{h \rightarrow 0} \frac{3(1+h)^2 - 4(1+h) - 5}{1} \quad [\text{by L' Hospital's rule}] \\
 &= -1
 \end{aligned}$$

$$\therefore Lf'(1) \neq Rf'(1)$$

$\Rightarrow f(x)$ is differentiable at $x = 1$. Also,

$$f(-2) = -8 - 8 + 10 + 6 = 0$$

$$f(3) = 27 - 18 - 15 + 6 = 0$$

$$f(-2) = f(3)$$

Thus, $f(x)$ satisfies all the conditions of Rolle's theorem.

There exists $c \in [-2, 3]$ such that

$$f'(c) = 0$$

$$\Rightarrow (c-1)(2c^2 - 3c + 1) = 0$$

$$\Rightarrow c = 1 \text{ or } 2c^2 - 3c + 1 = 0$$

$$\Rightarrow (c-1)(2c-1) = 0$$

$$\therefore c = 1 \text{ or } 1/2$$

Q17 (1) - Mean Value Theorem

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

We have, $f(x) = 2x^2 + 3x + 5$

Since, $f(x)$ satisfies all the conditions of LMVT at $x = 2$.

$$\therefore f'(2) = \frac{f(a)-f(1)}{a-1}$$

$$\Rightarrow 11 = \frac{2a^2+3a+5-10}{a-1}$$

$$\Rightarrow 11a - 11 = 2a^2 + 3a + 5 - 10$$

$$\Rightarrow 2a^2 - 8a + 6 = 0$$

$$\Rightarrow a^2 - 4a + 3 = 0$$

$$\Rightarrow (a-3)(a-1) = 0$$

$$\Rightarrow a = 1 \text{ or } 3$$

$$\Rightarrow a = 3$$

Q18 (4) - Mean Value Theorem

Given, $f(x) = 8x^2 - 7x + 5$

$$\begin{aligned} \therefore f'(c) &= 16c - 7 = \frac{f(6) - f(-6)}{12} \\ &= \frac{(8 \times 36 - 7 \times 6 + 5) - (8 \times 36 + 7 \times 6 + 5)}{12} = -7 \\ &\Rightarrow c = 0 \end{aligned}$$

Q19 (2) - Mean Value Theorem

It is given that f is continuous and differentiable for all real x .

$$\begin{aligned} \frac{f(4)-f(2)}{4-2} &= f'(c) \Rightarrow \frac{f(4)-f(2)}{2} \geq 6 \\ \Rightarrow \frac{f(4)+4}{2} &\geq 6 \Rightarrow f(4) + 4 \geq 12 \end{aligned}$$

$$\Rightarrow f(4) \geq 8$$

Q20 (2) - Mean Value Theorem

We have, $f(x) = x^3 + 7x - 1, x \in [0, 1]$

$$\therefore f(0) = -1$$

$$f(1) = 7$$

$$f(0) \neq f(1)$$

By LMVT, we have

$$\frac{f(1)-f(0)}{1-0} = 3c^2 + 7$$

Hints & Solutions

#MathBoleTohMathonGo

$$\Rightarrow 3c^2 + 7 = 8 \Rightarrow 3c^2 = 1$$

$$\Rightarrow c^2 = \frac{1}{3} \Rightarrow c = \pm \frac{1}{\sqrt{3}}$$

$$\therefore c = \frac{1}{\sqrt{3}} \in (0, 1)$$

Q21 (3) - Mean Value Theorem

Given, $f(x) = |1 - x|, 1 \leq x \leq 2$

$$\Rightarrow f(x) = -(1 - x), x \in [1, 2]$$

$$f(1) = 0$$

$$f(2) = 1$$

$$f(1) \neq f(2)$$

By LMVT, we have

$$\frac{f(2)-f(1)}{2-1} = f'(c), \forall c \in (1, 2) \Rightarrow \frac{1-0}{1} = f'(c)$$

$$\Rightarrow f'(c) = 1, \forall c \in (1, 2)$$

and $g(x) = -(1 - x) + b \sin\left(\frac{\pi}{2}x\right) = (x - 1) + b \sin\left(\frac{\pi}{2}x\right)$

$$\Rightarrow g(1) = b \Rightarrow g(2) = 1$$

If Rolle's theorem applicable to g , then $b = 1$.

Q22 (3) - Mean Value Theorem

Given that, $P(x) = 51x^{101} - 2323x^{100} - 45x + 1035$

Using Rolle's theorem, we consider anti-derivative of $P(x)$

i.e. consider $F(x) = \frac{x^{102}}{2} - \frac{2323x^{101}}{101} - \frac{45x^2}{2} + 1035x + c$

Then, being a polynomial function $F(x)$ is continuous and differentiable.

$$\text{Now, } F\left(45^{1/100}\right) = \frac{(45)^{\frac{102}{100}}}{2} - \frac{2323(45)^{\frac{101}{100}}}{101} - \frac{45(45)^{\frac{2}{100}}}{2} + 1035(45)^{\frac{1}{100}} + C$$

$$= \frac{45}{2}(45)^{\frac{2}{100}} - 23 \times 45(45)^{\frac{1}{100}} - \frac{45(45)^{\frac{2}{100}}}{2} + 1035(45)^{\frac{1}{100}} + C$$

$$= C$$

$$\text{and } F(46) = \frac{(46)^{102}}{2} - \frac{2323(46)^{101}}{101} - \frac{45(46)^2}{2} + 1035(46) + c$$

$$= 23(46)^{101} - 23(46)^{101} - 23 \times 45 \times 46 + 1035 \times 46 = c$$

$$\therefore F\left(45^{\frac{1}{100}}\right) = F(46) = c$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$\therefore$ Rolle's theorem is applicable. Hence, there must exist atleast one root of $F'(x) = 0$.

i.e. $P(x) = 0$ in the interval $(45^{1/100}, 46)$.

Q23 (1) - Monotonicity

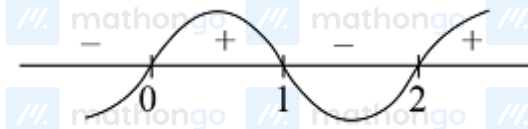
We have, $f(x) = x^2(x - 2)^2$

$$\Rightarrow f'(x) = 2x(x - 2)^2 + x^2 \cdot 2(x - 2)$$

$$= 2x[(x - 2)^2 + x^2 - 2x]$$

$$= 2x(x^2 + 4 - 4x + x^2 - 2x)$$

$$= 2 \times (2x^2 - 6x + 4) = 4x^3 - 12x^2 + 8x$$



For $f(x)$ to be increasing, we must have

$$f'(x) > 0$$

$$4x^3 - 12x^2 + 8x > 0$$

$$4x(x - 2)(x - 1) > 0$$

$$\therefore x \in (0, 1) \cup (2, \infty)$$

Q24 (1) - Monotonicity

We have, $f(x) = x^3 + 6x^2 + (9 + 2k)x + 1$

It is given that, $f(x)$ is increasing.

$$\therefore f'(x) \geq 0 \Rightarrow 3x^2 + 12x + (9 + 2k) \geq 0$$

$$\text{If } D = (12)^2 - 4 \times 3 \times (9 + 2k) \leq 0 \quad \left[\begin{array}{l} \because ax^2 + bx + c \geq 0 \\ \text{if } a > 0 \text{ and } D \leq 0 \end{array} \right]$$

$$\Rightarrow 144 - 108 - 24k \leq 0$$

$$\Rightarrow 36 - 24k \leq 0$$

$$\therefore k \geq 3/2$$

Q25 (3) - Monotonicity

Hints & Solutions

#MathBoleTohMathonGo

Given, $f(x) = \int \left(2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{1+x^2}} \right) dx$

$$\Rightarrow f'(x) = 2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{1+x^2}}$$

$$f'(x) = \frac{2(1+x^2) - 1 - \sqrt{1+x^2}}{(1+x^2)}$$

Clearly, $1+x^2 > 0, \forall x$

For the sign of numerator, Let $\phi(x) = 2(1+x^2) - 1 - \sqrt{1+x^2}$

$$\phi(x) = 2 + 2x^2 - 1 - \sqrt{1+x^2} = 2x^2 - \sqrt{1+x^2} + 1$$

$$\phi(x) = 2x^2 + 1 - \sqrt{1+x^2}$$

Now, $2x^2 > x^2, \forall x$

$$2x^2 + 1 > x^2 + 1, \forall x$$

$$\sqrt{2x^2 + 1} > \sqrt{x^2 + 1}, \forall x$$

$$\text{and } 2x^2 + 1 > \sqrt{2x^2 + 1}, \forall x$$

$$\Rightarrow 2x^2 + 1 > \sqrt{x^2 + 1}, \forall x$$

$$\Rightarrow 2x^2 + 1 - \sqrt{x^2 + 1} > 0, \forall x$$

$$\Rightarrow \phi(x) > 0, \forall x$$

Thus, $f'(x) > 0, \forall x$,

i.e. $x \in (-\infty, \infty)$

Hence, $f(x)$ is an increasing function.

Q26 (2) - Monotonicity

(1) Let $f(x) = e^x - 1 - x$

$$\Rightarrow f'(x) = e^x - 1 > 0, \forall x \in (0, 1)$$

So, $f(x)$ is increasing, when $0 < x < 1$

$$\Rightarrow f(x) > f(0) \text{ or } e^x - 1 - x > 0$$

$$\Rightarrow e^x > 1 + x$$

(2) Let $g(x) = \log_e(1+x) - x$

$$\Rightarrow g'(x) = \frac{1}{1+x} - 1 = \frac{-x}{1+x} < 0, \forall x \in (0, 1)$$

So, $g(x)$ is decreasing, when $0 < x < 1$

$$\Rightarrow g(0) > g(x) \Rightarrow \log_e(1+x) < x$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

(3) Let $h(x) = \sin x - x$

$\Rightarrow h'(x) = \cos x - 1 < 0, \forall x \in (0, 1)$

So, $h(x)$ is decreasing, when $0 < x < 1$

$\Rightarrow h(x) < h(0)$

$\therefore \sin x < x$

(4) Let $g(x) = \log_e x - x$

$\Rightarrow g'(x) = \frac{1}{x} - 1$

$\therefore g'(x) > 0, \forall x \in (0, 1)$

or $g(x) < g(1)$

$\Rightarrow \log_e x - x < -1 \Rightarrow x - 1 > \log_e x$

$\Rightarrow x > \log_e x$

Q27 (1) - Monotonicity

We have, $f(x) = xe^{x(1-x)}$

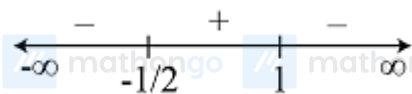
$\Rightarrow f'(x) = e^{x(1-x)} + x(1-2x)e^{x(1-x)}$

$\Rightarrow f'(x) = (1+x-2x^2)e^{x(1-x)}$

$\Rightarrow f'(x) = -(x-1)(2x+1)e^{x(1-x)}$

Since, $e^{x(1-x)} > 0$, for all x .

Therefore, sign of $f'(x)$ for different values of x are as shown below



Clearly, $f(x)$ is increasing on $[-1/2, 1]$ and decreasing on $(-\infty, -1/2] \cup [1, \infty)$.

Q28 (4) - Monotonicity

Given, $f(x) > x \dots (i)$

$\Rightarrow f[f(x)] > f(x) \dots (ii)$

From Eqs. (i) and (ii), we get

Hints & Solutions

#MathBoleTohMathonGo

$$\Rightarrow f[f(x)] > x$$

$$\Rightarrow f[f(x)] - x > 0$$

$\therefore$ No real root for $f[f(x)] = x$

Q29 (2) - Monotonicity

Given, $x = \frac{1}{1+t^2}, y = \frac{1}{t(1+t^2)}$

$$\Rightarrow \frac{dx}{dt} = \frac{-2t}{(1+t^2)^2}, \frac{dy}{dt} = \frac{-1+3t^2}{t^2(1+t^2)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1+3t^2}{2t^3}$$

$$\frac{dy}{dx} > 0, \text{ if } t > 0$$

$$\Rightarrow x = \frac{1}{1+t^2} \in (0, 1)$$

Hence, $f(x)$ is increasing in $(0, 1)$

Q30 (2) - Monotonicity

Given, $f(x) = \frac{\log(\pi+x)}{\log(e+x)}$

$$\Rightarrow f'(x) = \frac{\log(e+x) \times \frac{1}{\pi+x} - \log(\pi+x) \times \frac{1}{e+x}}{[\log(e+x)]^2}$$

$$= \frac{(e+x)\log(e+x) - (\pi+x)\log(\pi+x)}{(e+x)(\pi+x)[\log(e+x)]^2}$$

$$= \frac{e\log(e+x) + x\log(e+x) - \pi\log(\pi+x)}{(e+x)(\pi+x)[\log(e+x)]^2}$$

$$= \frac{x\log\left(\frac{e+x}{\pi+x}\right) + [e\log(e+x) - \pi\log(\pi+x)]}{(e+x)(\pi+x)[\log(e+x)]^2}$$

Now, $\pi+x > e+x \Rightarrow \log\left(\frac{e+x}{\pi+x}\right) < 0$

$$\Rightarrow f'(x) < 0$$

and also $\log(e+x) < \pi\log(\pi+x) \Rightarrow f'(x) < 0$

Thus, $f(x)$ is decreasing on $(0, \infty)$.

Q31 (4) - Monotonicity

$$h(x) = \frac{\log(f(x) \cdot g(x))}{\log(a)} = \log a^{\{a^{|x|} \operatorname{sgn} x\} + [a^{|x|} \operatorname{sgn} x]}$$

$$= \frac{\log a^{a^{|x|} \operatorname{sgn} x}}{\log a} = a^{|x|} \operatorname{Sgn} x \cdot \frac{\log a}{\log a}$$

$$= a^{|x|} \operatorname{sgn}$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo



$\therefore h(x)$ is odd and increasing for $a > 1$.

Q32 (2) - Monotonicity

Given, $f(x) = x^3 + ax^2 + bx + 5 \sin^2 x$ is increasing on R .

$$\Rightarrow f'(x) > 0, \forall x \in R$$

$$\Rightarrow 3x^2 + 2ax + b + 5 \sin 2x > 0, \forall x \in R$$

$$\Rightarrow 3x^2 + 2ax + (b - 5) > 0, \forall x \in R$$

$$[\because \sin 2x \geq -1]$$

$$\text{NoW, } (2a)^2 - 4 \times 3 \times (b - 5) < 0$$

$$\left[\begin{array}{l} \because \text{ if } f(x) = ax^2 + bx + c > 0 \\ \text{and } a > 0 \Rightarrow D < 0 \end{array} \right]$$

$$\Rightarrow 4a^2 - 12(b - 5) < 0$$

$$\therefore a^2 - 3b + 15 < 0$$

$$\text{Now, } (2a)^2 - 4 \times 3 \times (b - 5) < 0$$

$$\left[\begin{array}{l} \because \text{ if } f(x) = ax^2 + bx + c > 0 \\ \text{and } a > 0 \Rightarrow D < 0 \end{array} \right]$$

$$\Rightarrow 4a^2 - 12(b - 5) < 0$$

$$\therefore a^2 - 3b + 15 < 0$$

Q33 (4) - Maxima Minima

Since, $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$

$$\Rightarrow f'(x) = 6x^2 - 18ax + 12a^2$$

For maxima or minima,

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$f'(x) = 0 \Rightarrow 6(x^2 - 3ax + 2a^2) = 0$$

$$\Rightarrow x^2 - 3ax + 2a^2 = 0$$

$$\Rightarrow x^2 - 2ax - ax + 2a^2 = 0$$

$$\Rightarrow x(x - 2a) - a(x - 2a) = 0$$

$$\Rightarrow (x - a)(x - 2a) = 0$$

$$\Rightarrow x = a, x = 2a$$

$$\text{Now, } f''(x) = 12x - 18a$$

$$\text{at } x = a$$

$$f''(x) = 12a - 18a = -6a$$

$\therefore f(x)$ will be maximum at $x = a$,

$$\text{i.e. } p = a$$

Again at $x = 2a$

$$f''(x) = 24a - 18a = 6a$$

$\therefore f(x)$ will be minimum at $x = 2a$,

$$\text{i.e. } q = 2a$$

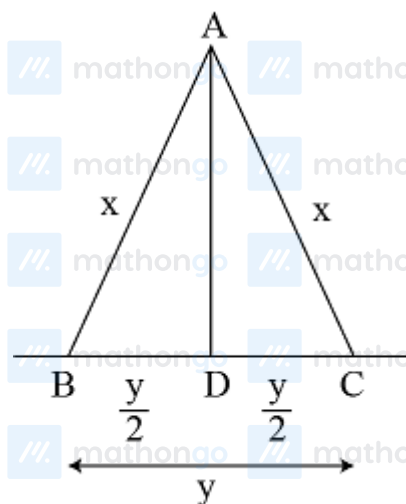
$$\text{Given, } p^2 = q$$

$$\Rightarrow a^2 = 2a$$

$$\therefore a = 2$$

Q34 (2) - Maxima Minima

Let ABC be a triangular park in which $AB = AC = x$ and $BC = y$



#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\text{Now, } AD = \sqrt{x^2 - \frac{y^2}{4}}$$

$$= \frac{1}{2} \sqrt{4x^2 - y^2}$$

$$\text{Area of } \triangle ABC (A) = \frac{1}{2} \times BC \times AD$$

$$A = \frac{1}{2} \times y \times \frac{1}{2} \sqrt{4x^2 - y^2}$$

$$A = \frac{1}{4} y \sqrt{4x^2 - y^2}$$

$$\frac{dA}{dy} = \frac{1}{4} \sqrt{4x^2 - y^2} + \frac{y}{4} \frac{1}{2\sqrt{4x^2 - y^2}} \times (-2y)$$

$$= \frac{\sqrt{4x^2 - y^2}}{4} - \frac{y^2}{4\sqrt{4x^2 - y^2}}$$

$$= \frac{4x^2 - y^2 - y^2}{4\sqrt{4x^2 - y^2}} = \frac{4x^2 - 2y^2}{4\sqrt{4x^2 - y^2}}$$

For maximum and minimum,

$$\frac{dA}{dy} = 0$$

$$\Rightarrow 4x^2 - 2y^2 = 0$$

$$\Rightarrow y^2 = 2x^2$$

$$\Rightarrow y = \sqrt{2}x$$

Now,

$$\frac{d^2A}{dy^2} = \frac{4\sqrt{4x^2 - y^2}(-4y) - (4x^2 - 2y^2) \frac{4}{2\sqrt{4x^2 - y^2}}(-2y)}{16(4x^2 - y^2)}$$

$$= \frac{-16y(4x^2 - y^2) + 4y(4x^2 - 2y^2)}{16(4x^2 - y^2)^{3/2}}$$

$$= \frac{-12y(4x^2 - y^2)}{16(4x^2 - y^2)^{3/2}} = \frac{-3y(4x^2 - y^2)}{4(4x^2 - y^2)^{3/2}}$$

$$= \frac{-3y}{4(4x^2 - y^2)^{1/2}}$$

$$\text{Then, } \left(\frac{d^2A}{dy^2} \right)_{\text{at } y=\sqrt{2}x} = \frac{-3\sqrt{2}x}{4(x^2 - 2x^2)^{1/2}}$$

$$= \frac{-3\sqrt{2}x}{4\sqrt{2}x} = -3/4 < 0$$

$\therefore$ Area is maximum, when $y = \sqrt{2}x$.

$$\text{and maximum} = \frac{1}{4} \sqrt{2}x \sqrt{2}x = \frac{1}{2} x^2$$

Hints & Solutions

#MathBoleTohMathonGo

Q35 (4) - Maxima Minima

We have,

$$P(x) = x^4 + ax^3 + bx^2 + cx + d$$

$$\Rightarrow P'(x) = 4x^3 + 3ax^2 + 2bx + c$$

It is given that, $x = 0$ is a root of $P'(x) = 0$.

$$\therefore P'(0) = 0 \Rightarrow c = 0$$

On putting $c = 0$, we get

$$P'(x) = x(4x^2 + 3ax + 2b)$$

Since, $x = 0$ is the only real root of $P'(x) = 0$

$$\therefore 4x^2 + 3ax + 2b = 0 \text{ has no real root.}$$

$$\Rightarrow 9a^2 - 32b < 0$$

It is given that, $P(-1) < P(1)$

$$\Rightarrow 1 - a + b + c < 1 + a + b + c + d$$

$$\Rightarrow a > 0$$

$$\text{But } 9a^2 - 32b < 0$$

$$\therefore b > 0$$

$$\text{Now, } P'(x) = x(4x^2 + 3ax + 2b) > 0, \forall x \in (0, 1)$$

 $\therefore P(x)$ is increasing in $(0, 1]$. $\Rightarrow P(1)$ is the maximum value of $P(x)$. Also,

$$P'(x) = x(4x^2 + 3ax + 2b) < 0, \forall x \in [-1, 0)$$

$$[\because 4x^2 + 3ax + 2b > 0, \forall x]$$

 $\Rightarrow P(x)$ is decreasing in $(-1, 0)$. $\Rightarrow P(-1)$ is not the minimum value of P .

Q36 (4) - Maxima Minima

$$\text{Let } f(x) = 2x^3 + 3x + k$$

Since, $f(x)$ has two distinct real roots $\in (0, 1)$ Then, $f'(x)$ must be zero for some $x \in (0, 1)$

$$\text{Here, } f'(x) = 6x^2 + 3x = 3x(2x + 1) = 0$$

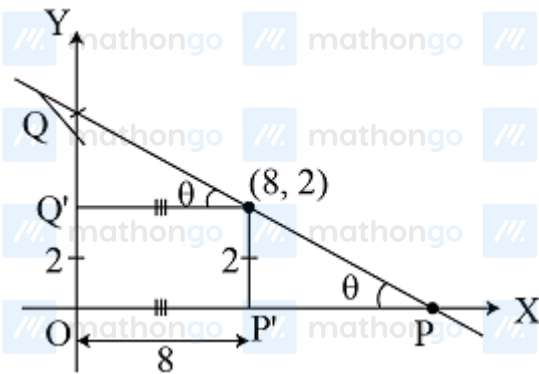
$$\Rightarrow x = 0, -\frac{1}{2} \notin (0, 1)$$

 $\therefore$ No value of k i.e. k does not exist.

Hints & Solutions

#MathBoleTohMathonGo

Q37 (4) - Maxima Minima



Here, $OP = OP' + PP' = 8 + 2 \cot \theta$

$$OQ = OQ' + QQ' = 2 + 8 \tan \theta$$

$$\therefore OP + OQ = 10 + 2(\cot \theta + 4 \tan \theta) \quad \therefore (i)$$

As, $AM \geq GM$

$$\frac{\cot \theta + 4 \tan \theta}{2} \geq \sqrt{\cot \theta \times 4 \tan \theta}$$

$$\Rightarrow \cot \theta + 4 \tan \theta \geq 4 \quad \therefore (ii)$$

From Eqs. (i) and (ii), we get

$$(OP + OQ)_{\min} = 10 + 8 = 18$$

Q38 (4) - Maxima Minima

$$\text{Given, } f(x) = e^{x^2} + e^{-x^2}, \quad g(x) = xe^{x^2} + e^{-x^2}$$

$$\text{and } h(x) = x^2e^{x^2} + e^{-x^2}$$

For any $x \in [0, 1]$, we have

$$x^2 \leq x \leq 1$$

$$\Rightarrow x^2e^{x^2} \leq xe^{x^2} \leq e^{x^2}$$

$$\Rightarrow e^{-x^2} + x^2e^{x^2} \leq e^{-x^2} + xe^{x^2} \leq e^{-x^2} + e^{x^2}$$

$$\Rightarrow h(x) \leq g(x) \leq f(x)$$

$$\text{Now, } f'(x) = 2x(e^{x^2} - e^{-x^2}) > 0, \forall x \in (0, 1]$$

$$\Rightarrow f(x) \text{ is increasing on } (0, 1].$$

$$\Rightarrow f(1) \text{ is maximum value of } f(x) \text{ on } [0, 1].$$

$$\Rightarrow 9 + e + e^{-1}$$

Hints & Solutions

#MathBoleTohMathonGo

$$\text{Also, } f(1) = g(1) = h(1) = e + e^{-1}$$

$$a = b = c = e + e^{-1}$$

Q39 (2) - Maxima Minima

$$\text{We have, } f(x) = 3x^4 - 4x^3 + 6x^2 + ax + b$$

$$\Rightarrow f'(x) = 12x^3 - 12x^2 + 12x + a$$

$$\Rightarrow f'(x) = 36x^2 - 24x + 12$$

$$= 12(3x^2 - 2x + 1) > 0, \forall x.$$

$\therefore f(x)$ has only one real root.

Since, $f'(x)$ changes in sign only once.

Therefore, only one extremum possible.

Q40 (1) - Maxima Minima

$$\text{It is given that, } f(x) = \begin{cases} e^x, & 0 \leq x \leq 1 \\ 2 - e^x, & 1 < x \leq 2 \\ x - e, & 2 < x \leq 3 \end{cases}$$

$$\text{and } g(x) = \int_0^x f(t) dt, x \in [0, 3]$$

$$g'(x) = f(x) = \begin{cases} 2 - e^{x-1}, & 0 < x \leq 2 \\ x - e, & 2 < x \leq 3 \end{cases}$$

$$\text{So, } g'(x) = 0$$

$$\Rightarrow x = 1 + \log_e 2 \text{ and } x = e$$

$$\text{Also, } g'(x) > 0 \text{ for } x \in (0, 1 + \log_e 2)$$

$$\text{and } g'(x) < 0 \text{ for } (1 + \log_e 2, 2)$$

$$\text{So, } g(x) \text{ attains a local maxima at } x = 1 + \log_e 2$$

$$\text{Similarly, } g'(x) < 0 \text{ for } 2 < x < e \text{ and } g'(x) > 0 \text{ for } e < x < 3 \Rightarrow g(x) \text{ attains local minimum at } x = e.$$